

Realistic Time-Domain Synthesis of Gravitational-Wave Detector Glitches using Class-Conditional Derivative Generative Adversarial Networks

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Gravitational-wave detectors such as LIGO, Virgo, and KAGRA are highly sensitive instruments susceptible to numerous noise sources. Short-duration transient noise events, known as glitches, pose a particular challenge for data analysis pipelines as they can mimic or obscure astrophysical signals. We present GlitchGAN, a class-conditional generative model, built upon the Conditional Derivative GAN (cDVGAN) architecture, that is capable of synthesizing realistic glitches directly in the time domain. The model is trained on high-quality reconstructions of seven common glitch types observed during LIGO’s third observing run (O3): Blip, Fast Scattering, Koi Fish, Low-Frequency Burst, Scattered Light, Tomte, and Whistle. We show that GlitchGAN generalizes effectively, learning to reproduce a diverse and physically consistent glitch space directly from these reconstructions. Moreover, because the model is conditioned on glitch class, it can generate *hybrid* or transitional glitch morphologies by interpolating across the class-conditioning vector after training. Synthetic glitches are validated against real glitches using the Gravity Spy classifier, widely used in the GW community for glitch classification, and an unsupervised analysis using UMAP embeddings. Gravity Spy classifies the majority of GlitchGAN’s synthetic glitches as the correct class while the UMAP analysis shows substantial overlap between real and synthetic samples in the reduced latent space. We further highlight a critical limitation of magnitude-only spectrograms: classifiers operating on magnitude Q -transforms can confidently misclassify physically unrealistic glitches from less robust models, underscoring the need for complementary validation methods that preserve phase information.

I. INTRODUCTION

The detection of gravitational waves (GWs) by the LIGO [1] and Virgo [2] detectors has opened a new observational window on the universe, enabling measurements of compact binary coalescences and other cataclysmic astrophysical events [3]. The extreme sensitivity required for these detections, however, renders the instruments susceptible to instrumental and environmental noise sources [4]. Among these, short-duration transient noise artefacts known as *glitches* [5–7] represent a persistent challenge for data analysis pipelines and extensive work is required to mitigate their impact, ensuring the quality of gravitational-wave data [8]. Glitches are unmodelled noise events that exhibit diverse morphologies—ranging from impulsive millisecond-duration events to extended, dispersive features spanning several seconds—and can mimic or obscure astrophysical signals [9–11]. When a glitch overlaps with a GW signal it can significantly bias parameter estimation [12–14], and approximately 25 of the ~ 90 confident detections up to the end of O3 required some form of glitch mitigation [3, 12, 15–17]. The contamination of astrophysical signals by glitches will be exacerbated in next-generation. Detectors such as Einstein Telescope [18] and Cosmic Explorer [19] are expected to detect on the order of $10^4 - 10^5$ binary black hole and neutron star mergers per year [20, 21], thus giving rise to more frequent glitch-signal overlaps.

The ability to synthesize realistic glitch populations is therefore of growing importance. Such capability enhances the realism of detector simulations and supports software injection campaigns [6, 22, 23], enabling more rigorous validation of GW detection pipelines and parameter estimation algorithms. Realistic synthetic glitches are also essential for the construction of mock data challenges (MDCs) and for data augmentation in machine learning-based glitch classification systems. A possible approach is to model glitches mathematically or using machine learning, given *a priori* knowledge of their production mechanism [24, 25]. Other works have developed models that mimic their principal characteristics using a few parameters [26]. However, such approaches either require prior knowledge of the glitch’s origin, which isn’t always available, or are limited to a restricted parametric form, making them difficult to scale across the full diversity of glitch morphologies.

Recently, generative models have been applied broadly within GW data analysis to address multiple challenges [27, 28]. For glitch synthesis, Generative Adversarial Networks (GANs) [29] have been used to augment spectrogram-based datasets for downstream classification tasks [30, 31]. The *gengli* glitch generator [32, 33] demonstrated that a GAN trained on Blip glitches extracted from detector data using BayesWave [34], the state-of-the-art method for reconstructing glitches from detector data, can successfully learn their underlying time-domain distribution, providing a useful framework for simula-

tions and MDCs. McGANn [35] implemented a conditional GAN capable of synthesizing five distinct waveform classes analogous to GW bursts in one model, with class interpolation enabling the generation of hybrid morphologies that blend features from the learned classes. Powell et al. [31] implemented generative modelling on a broader glitch class space by training separate GANs on magnitude Q -scan representations of each glitch class. However, extending this approach to a single model operating directly in the time domain remains unexplored, despite its advantages in preserving phase information and enabling direct use of time-series data in simulations and MDCs.

The principal obstacle to realistic time-domain glitch synthesis has been the availability of high-quality training data. BayesWave [34] is computationally expensive, and converges unreliably for certain glitch morphologies, making it impractical to construct the large-scale, diverse datasets that cover the broad glitch space. The recent development of DeepExtractor [36]—a deep learning framework for time-domain glitch reconstruction—offers fast, high-fidelity reconstructions that preserve both amplitude and phase information without imposing waveform smoothness assumptions, achieving a $10,000\times$ speed-up over BayesWave in realistic simulations.

In this paper, we introduce GlitchGAN, a class-conditional model trained on a dataset of DeepExtractor reconstructions of seven common glitch classes observed during LIGO’s O3 run: Blip, Fast Scattering, Koi Fish, Low-Frequency Burst, Scattered Light, Tomte, and Whistle. Examples of the glitches used for training and the corresponding GlitchGAN outputs are shown in Figure 1. GlitchGAN inherits the cDVGAN architecture [37, 38], which features an auxiliary discriminator applied to the time derivative of signals, providing additional regularisation that improves training stability and sample fidelity compared to standard GAN counterparts.

We validate the quality of the DeepExtractor training data using the Gravity Spy classifier [39–41] and unsupervised t-SNE [42] and UMAP embeddings [43], which have been used in multiple studies to characterize the morphological distribution of glitches in LIGO data [44–46]. The evaluation establishes a realistic baseline against which synthetic glitches can be assessed. Applying the same evaluation to GlitchGAN’s outputs, we show that the model captures much of the morphological diversity across all seven classes, with substantial overlap between real and generated distributions in the t-SNE and UMAP latent spaces and with the majority of synthetic glitches correctly identified by Gravity Spy. We also demonstrate that GlitchGAN can interpolate between classes, producing novel hybrid morphologies that mix features from the learned classes. We further highlight a fundamental limitation of magnitude-only time-frequency representations as validation tools: when applied to physically unrealistic glitches from a diffusion-based comparison model [47], Gravity Spy assigns correct class labels with high confidence despite clear inconsistency in the time domain,

because the Q -transform discards phase information entirely. This underscores the necessity of complementary, phase-preserving validation methods.

The remainder of the paper is organized as follows: Section II describes the preparation of our training dataset. Section III details the GlitchGAN architecture and training procedure. Section IV presents results across multiple metrics. Finally, Section V discusses our findings and opportunities for future work.

II. DATA

A. Data curation

We construct our training dataset using DeepExtractor reconstructions of real LIGO glitches. To ensure high-quality, morphologically representative samples, we apply strict selection criteria:

1. Gravity Spy classification confidence $\sigma_{\text{GS}} \geq 0.9$, ensuring that the reconstructed glitches are representative of well-characterized classes.
2. Signal-to-noise ratio (SNR) ≥ 15 (sufficient amplitude to minimize reconstruction artifacts)
3. Detectors: LIGO Hanford (H1) and Livingston (L1)
4. Observing runs: O3a and O3b

We include seven glitch classes in the training of GlitchGAN, namely; Blip, Fast Scattering, Koi Fish, Low-Frequency Burst, Scattered Light, Tomte, and Whistle.

Blip glitches have a characteristic morphology of a symmetric teardrop shape in time-frequency in the range [30, 250] Hz with short-durations, ~ 0.04 s [11]. They appear in both Hanford and Livingston, as well as Virgo and GEO 600 [48]. Due to their abundance and form, these glitches hinder both the unmodeled burst and modelled CBC searches, with particular emphasis on compact binaries with large total mass, highly asymmetric component masses, and spins anti-aligned with the orbital angular momentum [9].

Fast Scattering glitches appear as short-duration arches ($\sim 0.2 - 0.3$ s) in the frequency range [20, 60] Hz. These glitches are strongly correlated with ground motion in range [0.1 – 0.3] Hz and [1 – 6] Hz, which in turn is associated with thunderstorms and human activity near the detector. Note that this class was added during O3, and it is more abundant in Livingston than Hanford, due to differences in ground motion and detector sensitivity [49].

Koi Fish glitches get their name from their frontal resemblance to Koi Fishes. They are also similar to *Blips* but typically feature a higher SNR, spanning the frequency range of [20, 1000] Hz.

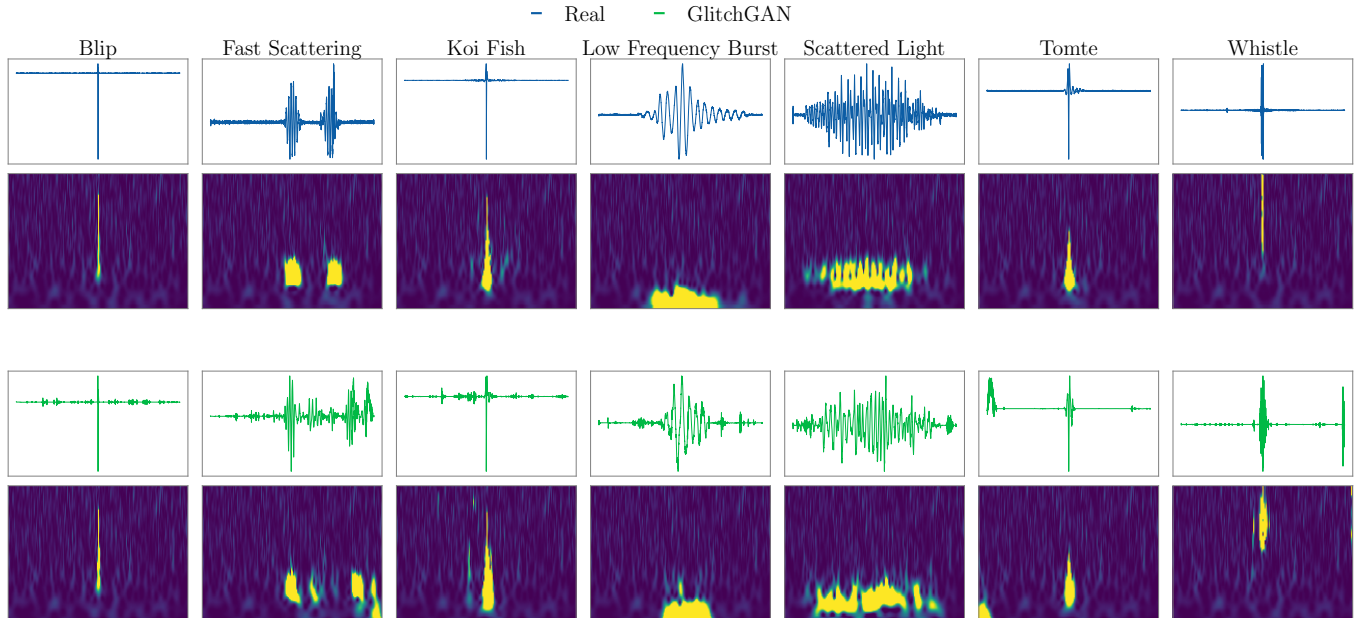


FIG. 1: Comparison of real and GlitchGAN-generated glitches across the seven glitch classes used for training, shown in both representations. **Row 1 (Real, time domain):** DeepExtractor reconstructions of real glitches. **Row 2 (Real, Q-transform):** Corresponding Q-scans of the same real glitches. **Row 3 (Synthetic, time domain):** Glitches generated by GlitchGAN. **Row 4 (Synthetic, Q-transform):** Corresponding Q-scans of the generated glitches. Each column shows one representative sample per class.

Low-Frequency Bursts are short-duration glitches (~ 0.25 s) in the frequency range $[10 - 20]$ Hz with a distinctive blob shape. These occurrences were prevalent in Livingston data during O1 and Hanford data in O3a.

Scattered Light glitches have longer duration harmonics ($\sim 2.0 - 4.0$ s), and in the time-frequency domain, they appear as arches being often stacked on top of each other. These glitches are problematic since their frequency content lies in the band of interest of GW astrophysical events. In O3, they were found to be coupled with the relative motion between the optical suspension system’s end test-mass chain and the reaction-mass chain [49].

Tomte glitches are also short-duration (~ 0.25 s) with characteristic triangular morphology.

Whistles have a characteristic V, U or W shape at higher frequencies with typical durations ~ 0.25 s. They are caused when radio-frequency signals beat with the voltage-controlled oscillators [50].

These seven classes collectively span wide time and frequency ranges, encompassing much of the morphological variability relevant to transient instrumental noise. While it is a common glitch class, we exclude the Extremely Loud class because its characterization is dominated by exceptionally high SNR rather than distinctive morphology, making it a poor target for generative modeling.

B. Dataset Preprocessing

We reconstruct with DeepExtractor a 2-second window centered on the glitch trigger time at a sampling rate of 4096 Hz, yielding 8192 samples per glitch waveform. Our goal was to obtain approximately 5,000 samples per class (1,250 per detector per observing run) under the selection criteria above. Unfortunately, data availability constraints limited acquisition for some classes. Fast Scattering yielded only 2,774 samples, and Low-Frequency Burst provided 4,316 samples. The total curated dataset contained 31,511 samples.

To mitigate training instabilities from class imbalance, we applied bootstrap resampling (sampling with replacement) to oversample all classes to 5,000 samples each, yielding a balanced training set of 35,000 total samples. Dataset composition by detector, observing run, and class is detailed in Table I. Before training, all reconstructed time series are normalized to the range $[-1, 1]$, followed by subtracting the mean. This normalization constrains the generative model to operate at a single effective SNR scale, but generated samples can be rescaled post-generation to any desired SNR for downstream applications.

TABLE I: Composition of the curated glitch dataset. Number of samples per class, detector (H1: Hanford, L1: Livingston), and observing run (O3a, O3b). Classes are bootstrapped to 5000 samples each for balanced training.

Glitch Class	H1		L1	
	O3a	O3b	O3a	O3b
Blip	1246	1247	1245	1243
Fast Scattering	143	143	1244	1244
Koi Fish	1247	1248	1243	1247
Low Freq. Burst	1246	1246	912	912
Scattered Light	1236	1243	1246	1248
Tomte	1031	1031	1247	1246
Whistle	1232	1232	1231	1232

III. METHODS

A. Model architecture: cDVGAN

Generative adversarial networks (GANs) [29], wherein a generator and discriminator are trained in an adversarial min-max game, have established themselves as a powerful framework for learning complex high-dimensional data distributions. Wasserstein GANs (WGANs) [51] improve training stability by replacing the Jensen–Shannon divergence with the Wasserstein-1 distance, and the gradient penalty variant (WGAN-GP) [52] further stabilizes optimization by enforcing a Lipschitz constraint on the discriminator. Conditional extensions [53] allow the generator to be steered towards a target class, enabling multi-class synthesis within a single model.

cDVGAN [37] is a conditional WGAN-GP that augments the standard two-player adversarial game with an auxiliary discriminator applied to the first-order time derivative of signals. Here we summarise the key components; full layer-by-layer architecture details are shown in Appendix A.

1. Training objective

Under the WGAN-GP framework, the loss for discriminator D is

$$L_D = \mathbb{E}_{\hat{x}}[D(\hat{x}, c)] - \mathbb{E}_x[D(x, c)] + \lambda \mathbb{E}_{\tilde{x}} \left[(\|\nabla_{\tilde{x}} D(\tilde{x}, c)\|_2 - 1)^2 \right], \quad (1)$$

where x and $\hat{x}$ are real and generated samples respectively, $\tilde{x}$ is a random interpolation between them, and $\lambda = 10$ is the gradient-penalty weight. The generator is updated to minimise

$$L_G = -\mathbb{E}_{\hat{x}}[D(\hat{x}, c)]. \quad (2)$$

In cDVGAN a second discriminator D_{∂} operates on the first-order derivative $\dot{x} = dx/dt$. The combined generator loss is

$$L_G^{\text{cDVGAN}} = -\frac{1}{2} \left(\mathbb{E}_{\hat{x}}[D(\hat{x}, c)] + \mathbb{E}_{\dot{\hat{x}}}[\dot{D}(\dot{\hat{x}}, c)] \right), \quad (3)$$

where both discriminators contribute equally. Each discriminator is updated five times per generator update.

2. Generator

The generator G maps a noise vector $z \sim \mathcal{N}(0, I_{100})$ and a one-hot class vector $c \in \mathbb{R}^7$ to a time series of length 8192. The class vector is first projected to a 32-dimensional embedding and concatenated with z , then passed through a dense layer and reshaped before five successive upsample-and-convolve blocks with batch normalisation and ReLU activations expand the sequence to the target length.

3. Discriminators

Both D and D_{∂} are 1D convolutional networks that map an input sequence and a class vector to a scalar critic value. Class conditioning uses projection [53]: a learned class embedding is dot-producted with the discriminator’s feature vector and added to the scalar output. D operates directly on the 8192-sample waveform; D_{∂} operates on the 8191-sample first-order difference sequence $\dot{x} = x_{t+1} - x_t$. The derivative discriminator provides additional regularisation by penalising unphysical high-frequency artefacts in generated waveforms [38].

B. Training procedure

The model was trained for 500 epochs with a batch size of 64 using the RMSprop optimiser and a learning rate of 10^{-4} for all network components. Training was performed on a NVIDIA A100 GPU (approximately 12 hours). Although training was conducted for the full 500 epochs, the final GlitchGAN configuration uses the model weights saved at epoch 210.

The selection of epoch 210 was based on a combination of qualitative and quantitative assessments. Generated glitches were inspected throughout training, and epoch 210 was found to produce the most realistic glitch morphologies. This choice is further supported by the generator loss evolution (Appendix B), which exhibits relatively stable behaviour around this epoch compared to other stages of training. Additionally, the UMAP analysis presented in Section IV B 2 shows improved overlap between the distributions of real and generated glitches at epoch 210 relative to models saved at epochs 100, 200, 300, 400, and 500.

No hyperparameter tuning was performed for this dataset; the architecture and hyperparameters are identical to those validated in [37], allowing us to assess out-of-the-box generalisation to this more complex and realistic glitch population. The implementation uses TensorFlow [54] and is publicly available at <https://github.com/tomdooney95/glitchgan>.

C. Comparison with a diffusion-based model

As a comparison against our cDVGAN architecture in GlitchGAN, we also trained another type of generative model on our dataset known as a latent Diffusion Transformer (DiT), which was adapted for 1D time series. Diffusion models [47, 55, 56] learn to generate data by explicitly modelling the process of iteratively adding and then removing noise from samples drawn from a simple prior distribution. They have become increasingly popular in recent years for their ability to generate realistic synthetic data. Full-resolution diffusion proved computationally prohibitive, so we first trained a convolutional autoencoder to compress inputs from (8192, 1) to a latent space of (256, 16), reducing computational cost by approximately three orders of magnitude while preserving essential morphological structure. The DiT was then trained in this latent space, with generated samples decoded back to the time domain for evaluation. This model serves primarily to illustrate the limitation of magnitude-only validation metrics, as discussed in Section IV.

D. Evaluation Methodology

We evaluate both the training data (DeepExtractor reconstructions) and GlitchGAN-generated glitches using two complementary approaches: classification with the Gravity Spy classifier and visualisation of low-dimensional embeddings via t-SNE and UMAP. For DeepExtractor, this establishes a quantitative baseline confirming the training data is morphologically representative; for GlitchGAN, it assesses how faithfully the model reproduces the underlying glitch distributions.

1. Gravity Spy Classification

We reconstruct with DeepExtractor 100 randomly selected glitches from the Gravity Spy dataset according to the criteria in Section II, drawing 25 samples per detector (H1, L1) per observing run (O3a, O3b). Since Gravity Spy classifies glitches surrounded by detector background noise i.e. $g(t) + n(t)$, each DeepExtractor reconstruction $\hat{g}(t)$ is injected into a segment of ‘quiet’ Hanford detector noise from O3 between 1262540000 and 1262540040 GPS time prior to reclassification (representing $n(t)$), using the `fetch_open_data` method from the package `gwpy` [57]. This stretch of data was visually inspected using Q -scans to confirm the absence of glitches or other significant noise features.

We follow a similar approach to evaluate the GlitchGAN-generated synthetic glitches, injecting them into the same noise segments before classification. The only difference is that we have to explicitly rescale the generated glitches to a fixed SNR to ensure they are

sufficiently loud for classification, since the generator is trained on normalized waveforms and does not learn an explicit SNR distribution. We fix the SNR of each glitch class according to the mean SNR of the respective class considering the selection criteria described in Section II. The mean SNR values used for each class are shown in Table II.

TABLE II: Mean SNR per glitch class used for software injections in the GlitchGAN evaluation.

Glitch Class	Mean SNR
Blip	29.8
Fast Scattering	36.4
Koi Fish	187.2
Low Frequency Burst	40.3
Scattered Light	31.5
Tomte	25.4
Whistle	27.1

2. Dimensionality Reduction

To visualise and compare the high-dimensional feature space of real and synthetic glitches, we use two complementary nonlinear dimensionality reduction methods: *t-distributed Stochastic Neighbor Embedding* (t-SNE) [42] and *Uniform Manifold Approximation and Projection* (UMAP) [58]. While *Principal Component Analysis* (PCA) captures global variance through linear projections, it cannot resolve the nonlinear structure present in morphologically diverse glitch populations; t-SNE and UMAP are instead designed to preserve local cluster structure in the embedding. t-SNE embeds data by minimising the Kullback–Leibler divergence between probability distributions encoding pairwise similarities in the original and reduced spaces; it excels at revealing local cluster structure but can distort global geometry. UMAP constructs a nearest-neighbour graph in the high-dimensional space and optimises a low-dimensional counterpart to preserve both local and global topological structure, scaling efficiently to the dataset sizes used here.

For DeepExtractor’s reconstructions, we aim to verify that the reconstructed glitches cluster according to their correct Gravity Spy class in an unsupervised manner, confirming that the reconstructions preserve the morphological characteristics recognized by Gravity Spy. For GlitchGAN-generated synthetic glitches, we evaluate whether they cluster with the real reconstructions of their respective classes, indicating that the model has successfully learned to reproduce the underlying morphological distribution of each class.

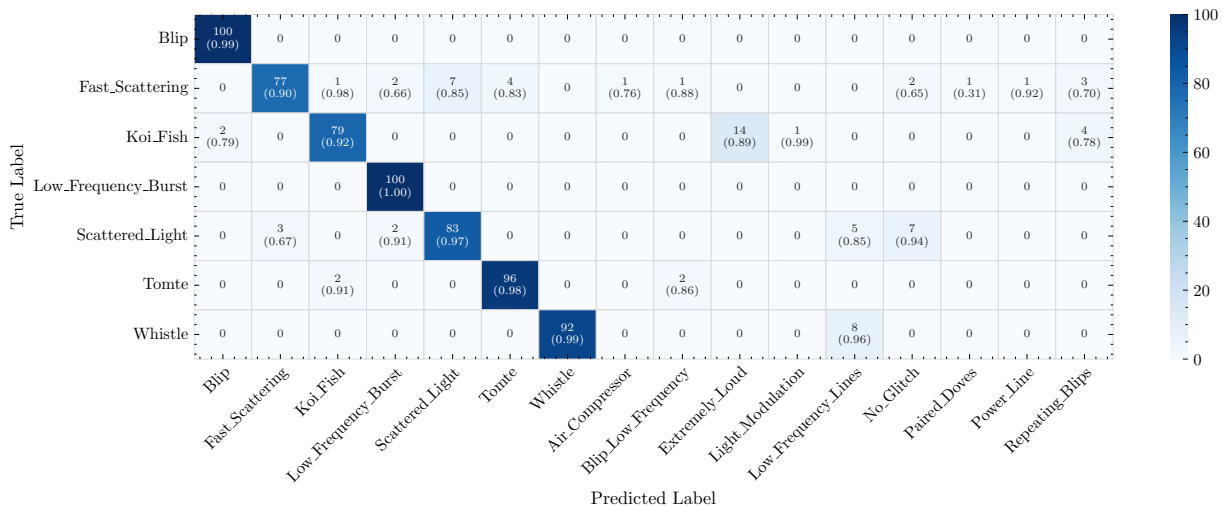


FIG. 2: Confusion matrix for Gravity Spy classification of 100 DeepExtractor reconstructions per class (700 total). Each cell shows the number of predictions with the mean Gravity Spy confidence in brackets.

IV. RESULTS

A. Validation of DeepExtractor training data

1. Classification evaluation using Gravity Spy

Tables III and IV summarize the results of classifying 700 DeepExtractor reconstructions (100 per class) with Gravity Spy. Since Gravity Spy supports additional classes beyond our seven training classes, the matrix is asymmetric. Of the 700 reconstructed glitches, 627 are correctly identified by Gravity Spy, corresponding to 89.57% overall accuracy. Correctly classified reconstructions achieve a mean confidence of 96.77%, confirming that DeepExtractor effectively preserves the morphological characteristics recognized by Gravity Spy. Misclassified samples show lower confidence of 84.84%, suggesting that classifier confidence may serve as a useful quality veto.

TABLE III: Summary of Gravity Spy results on DeepExtractor reconstructions (100 samples per class, 700 total).

Total samples	700
Correct predictions	627
Incorrect predictions	73
Overall accuracy	89.57 %
Mean confidence (correct)	96.77 %
Mean confidence (incorrect)	84.84 %

Table IV shows that class-level accuracy varies considerably. All Blip and Low-Frequency Burst glitches are correctly classified, while Fast Scattering and Koi Fish achieve the lowest accuracies at 77% and 79%, respectively. Figure 2 shows the full confusion matrix. Most

TABLE IV: Per-class correct prediction count and mean Gravity Spy confidence for DeepExtractor reconstructions.

True label	Correct count	Mean confidence (%)
Blip	100	99.30
Fast Scattering	77	89.78
Koi Fish	79	91.59
Low Frequency Burst	100	99.85
Scattered Light	83	96.95
Tomte	96	98.21
Whistle	92	99.33

misclassifications occur between morphologically similar classes: among the 21 misclassified Koi Fish glitches, 14 were labelled as Extremely Loud, consistent with their exceptionally high SNRs (220–1270) overwhelming finer morphological details. The 23 misclassified Fast Scattering samples are distributed across several related classes (Scattered Light, Tomte, Repeating Blips), reflecting the known morphological similarity between low-frequency scattering noise types. Overall, the misclassification patterns are morphologically explainable and do not indicate systematic failure of the reconstruction.

2. Unsupervised clustering with t-SNE and UMAP

Figure 3 shows three-dimensional t-SNE (left) and UMAP (right) embeddings of the DeepExtractor reconstructions, visualized from three complementary view-points. The reconstructions form well-separated clusters that align closely with the Gravity Spy labels, demonstrating that morphologically distinct glitch types occupy distinct regions of the time-domain latent space. Partial overlap between the Blip and Koi Fish clusters is visible, consistent with prior observations that these classes

may share a physical origin [59]. The good clustering structure validates that the 31,511 DeepExtractor reconstructions used to train GlitchGAN are morphologically representative of seven distinct classes and are not dominated by reconstruction artifacts or label noise.

B. Validation of GlitchGAN synthetic glitches

1. Classification evaluation using Gravity Spy

To quantify synthetic glitch realism we follow a similar approach to the DeepExtractor evaluation, injecting 100 generated glitches per class into quiet Hanford O3 background noise and classifying them with Gravity Spy. The resulting confusion matrix shown in Figure 4 reflects classifier predictions and confidence scores in brackets. The overall classification accuracy is $\sim 79.3\%$. While this is lower than the results for DeepExtractor reconstructions, the misclassifications are consistent with known limitations of Gravity Spy and the use of a fixed injected SNR according to the mean SNR of each glitch class.

Most misclassifications occur in the Koi Fish and Scattered Light classes. While only 70% of the generated Koi Fish samples were correctly classified, the majority of the remaining samples were misidentified as Extremely Loud, Light Modulation and Blip. This behaviour is consistent with known morphological similarities between Koi Fish and these glitch classes, which may arise from related physical mechanisms differing primarily in SNR [59]. Moreover, changing the injected SNR of the generated Koi Fish samples to 150 raises the classification accuracy to 79% with an average confidence of $\sim 95\%$, further illustrating the dependence of Gravity Spy classifications on the injected SNR.

This behaviour is further supported by injecting glitches at an SNR of 20. In this lower-SNR regime, none of the generated Koi Fish samples were correctly classified. Instead, they were misidentified as other morphologically similar classes. A comparable SNR dependence is observed for the Whistle class. Injecting the generated Whistle glitches at an SNR of 50 yields an accuracy of 94%, whereas injecting them at an SNR of 20 leads to frequent misclassification as No Glitch.

For Scattered Light, Gravity Spy often predicts Fast Scattering, which is expected given their related scattering origins at low frequency. This was also observed for DeepExtractor’s original reconstructions of both Fast Scattering and Scattered Light shown in Figure 2.

Another class of interest is the Whistle class, where 72% of glitches were correctly identified. Increasing the injected SNR from 27 to 50 raises the classification accuracy to 94%, illustrating another case of strong SNR dependence of Gravity Spy classifications for this class.

It is also important to note that some misclassifications may arise from stochastic interactions between the injected glitches and the specific noise realization into which they are injected, which can differ from the original

conditions of the glitch sample. Additionally, labelling uncertainties in the Gravity Spy training set may contribute, and the discarded phase information in Q -scans may play a role—an issue we discuss further below.

2. Global morphological similarity via UMAP

To assess whether GlitchGAN captures the overall structure of the real glitch distribution, we project both real and synthetic glitch samples into a three-dimensional latent space using UMAP [43].

Figure 5 shows two complementary perspectives of this embedding space, computed from 500 samples per class. Corresponding individual class-wise UMAP projections in Figure 6 provide additional detail. Strong overlap between real (blue) and synthetic (green) samples indicates that GlitchGAN has learned the general morphological structure of the different glitch types.

For most classes (Blip, Fast Scattering, Scattered Light, Tomte), synthetic and real samples show close clustering with substantial overlap. The Whistle class exhibits the most noticeable divergence, with synthetic samples forming a partially separate cluster, suggesting that GlitchGAN captures the dominant Whistle morphology but does not fully explore the full morphological diversity within this class. This is consistent with the relative complexity of the Whistle waveform structure and may reflect limitations of the fixed GAN architecture when applied to this particular class without hyperparameter tuning.

3. Generating hybrid glitches via class interpolation

A key advantage of class-conditional generative models is the ability to produce “hybrid” or transitional morphologies by sampling mixed class vectors.

We explore two sampling strategies:

1. **Simplex mixing:** The class vector is normalized such that all coefficients sum to 1, i.e., sampled from a $k = 7$ simplex. This constrains mixing proportions to sum to 100%.
2. **Uniform mixing:** Each class coefficient is independently sampled from $[0, 1]$ without normalization constraint, allowing more extreme interpolations.

Figure 7 shows glitch examples from both mixing strategies, demonstrating intermediate morphologies. As shown, samples from these datasets exhibit mixed morphological characteristics drawn from multiple glitch classes as per the sampled class vector. The UMAP projection of mixed samples in Figure 8 confirms that these hybrid samples smoothly interpolate between pure-class point clouds, with uniform mixing exploring the broadest region of the latent space.

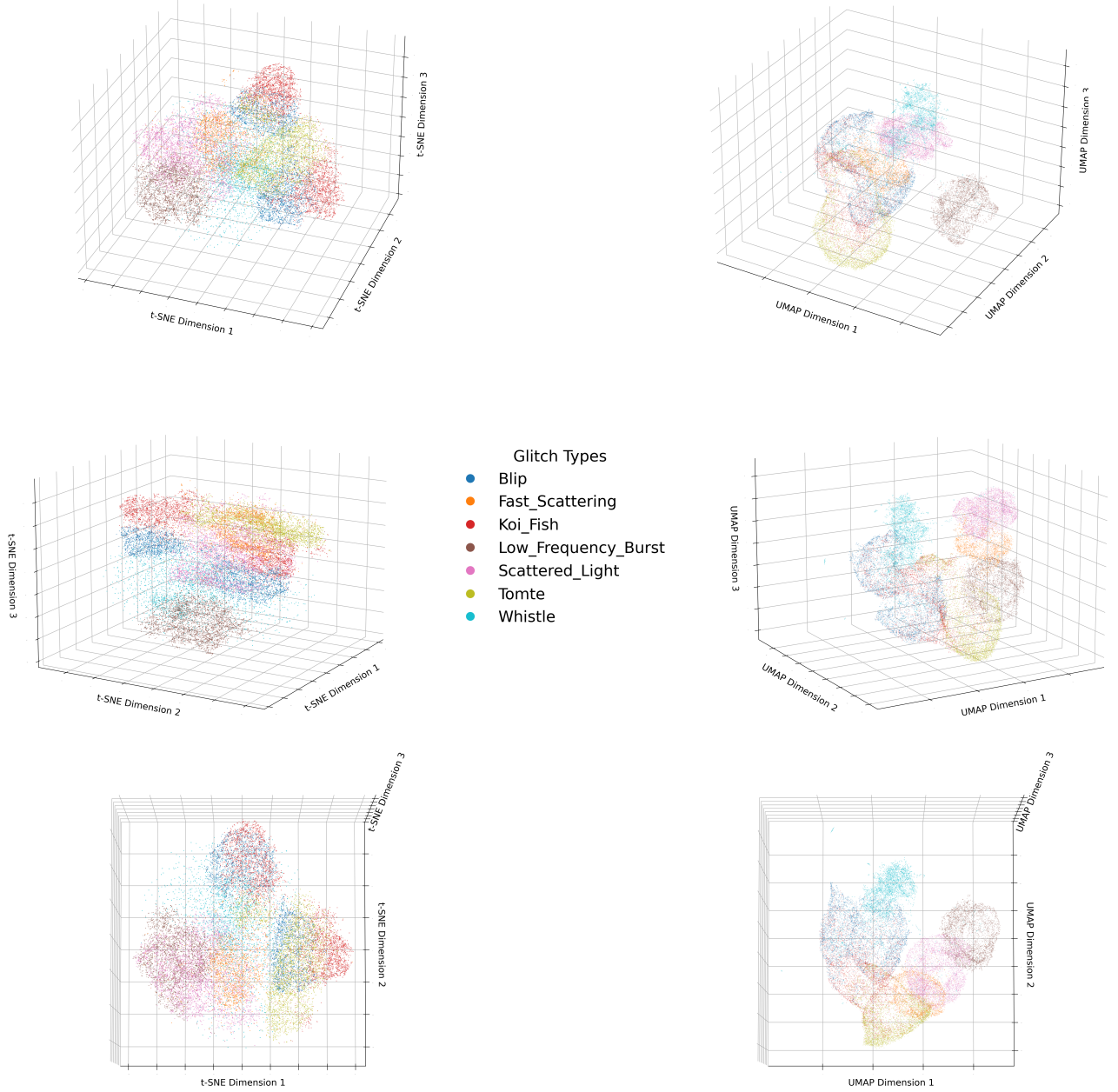


FIG. 3: Three-dimensional *t*-SNE (left) and UMAP (right) embeddings of DeepExtractor reconstructions across three viewpoints. Colours correspond to Gravity Spy glitch classes. Well-separated clusters demonstrate that reconstructions preserve class-specific morphological structure in the time domain. The partial overlap between Blip and Koi Fish (visible in the middle row) is consistent with their known morphological proximity.

While hybrid samples are not further analyzed in this study, they have potential applications for pipeline validation and data augmentation for glitch detection or classification systems.

C. Limitations of magnitude-only representations

We further evaluated the DiT model as a comparison; the full Gravity Spy and UMAP results are shown in Appendix D. DiT-generated samples overlap with real glitches primarily for low-frequency classes (e.g., Fast Scattering, Scattered Light), but show reduced diversity and separable clusters for most other classes where

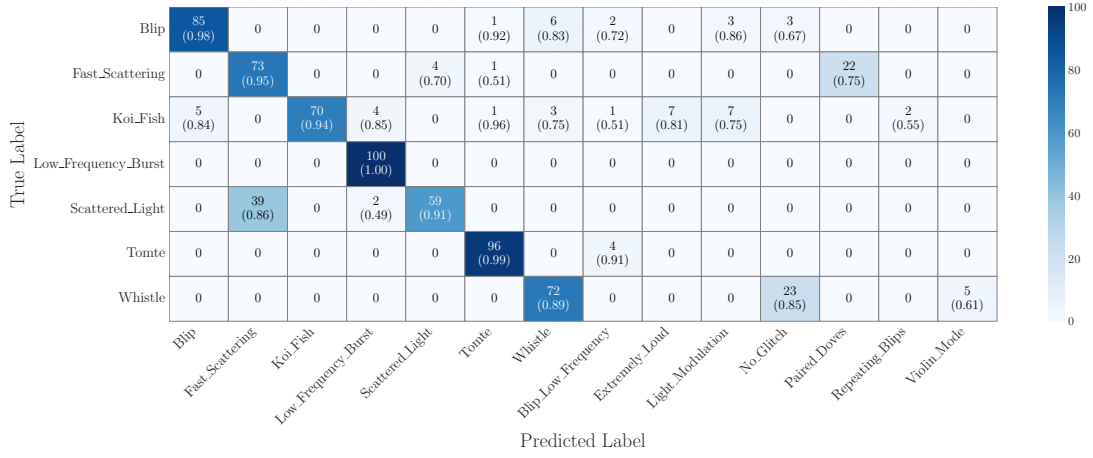


FIG. 4: Confusion matrix showing Gravity Spy classifications of glitches generated by GlitchGAN, with 100 samples per class. Each cell indicates the number of samples predicted for each class, with the average Gravity Spy confidence shown in brackets.

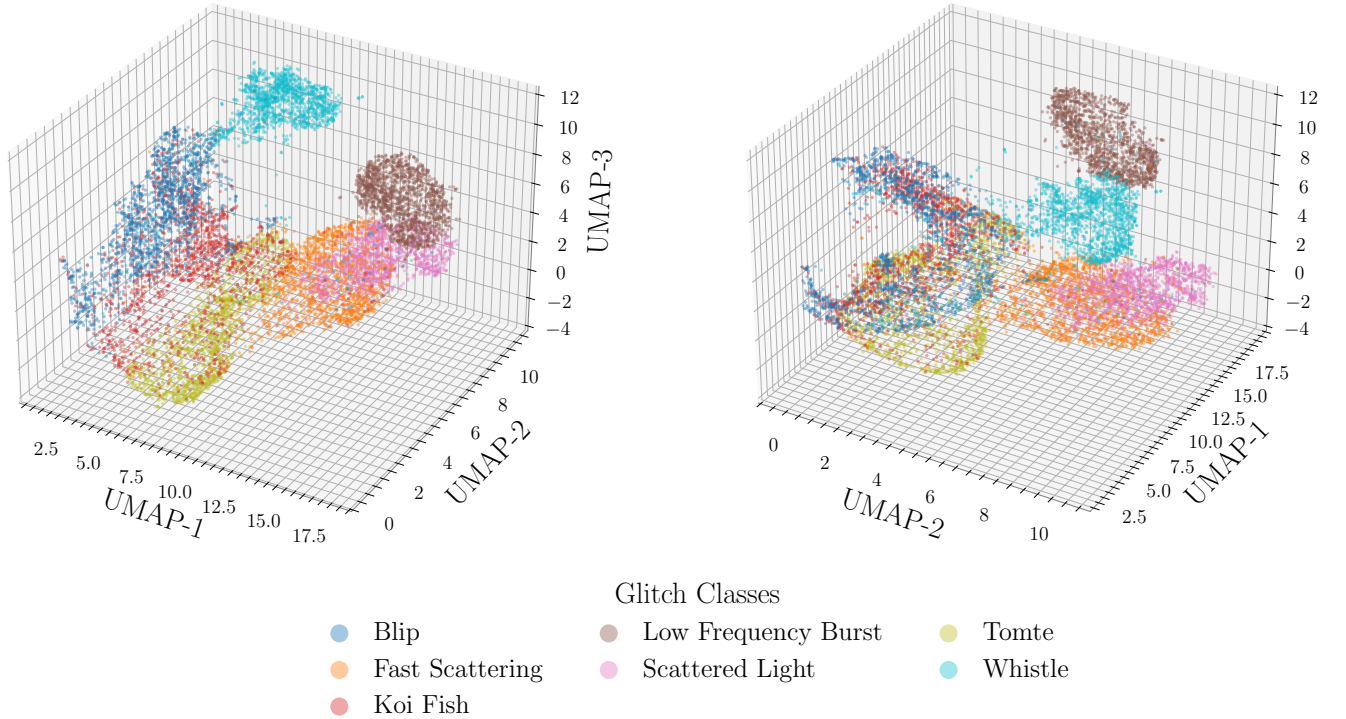


FIG. 5: Two complementary views of three-dimensional UMAP embeddings showing real and generated glitch samples. Each color corresponds to one of the seven glitch classes used during training.

GlitchGAN was more successful.

A notable example is the Low Frequency Burst class: UMAP reveals two clearly distinct clusters (real vs. synthetic), yet Gravity Spy classifies 87% of DiT-generated samples correctly with 97% mean confidence. Figure 9 illustrates why: while the time-domain waveforms of real and DiT-generated Low Frequency Bursts are clearly distinct, their magnitude Q -scan representations appear similar, explaining the classifier’s high confidence despite

the physically unrealistic time-domain morphology.

This finding underscores a critical limitation: classifiers operating on magnitude-only representations cannot reliably assess glitch realism. Complementary validation methods preserving phase information (such as UMAP on time-domain waveforms) are essential for ensuring synthetic glitches are suitable for detector simulations.

Overall, while the DiT model demonstrates that latent

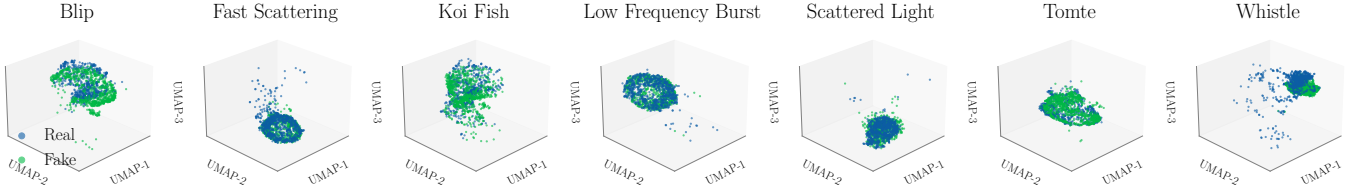


FIG. 6: Three-dimensional UMAP embeddings for each glitch class, comparing real (blue) and GlitchGAN-generated (green) samples. Each panel corresponds to one glitch type and is shown from a rotated viewpoint to highlight intra-class structure. The close overlap between real and synthetic distributions across most classes demonstrates that GlitchGAN effectively learns class-specific morphological features, though some variation remains for more complex glitch morphologies.

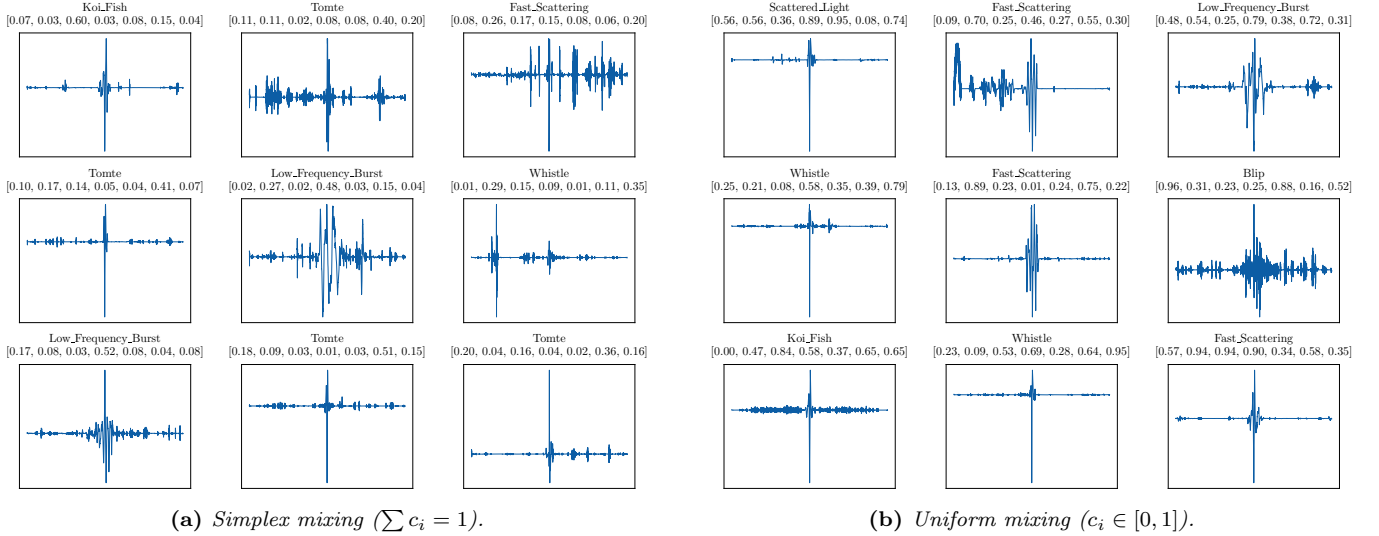


FIG. 7: Examples of hybrid glitch waveforms generated by GlitchGAN using two class-conditioning strategies. (a) Simplex mixing samples each class vector from a $k = 7$ simplex, ensuring coefficients sum to one. (b) Uniform mixing samples each coefficient independently from a uniform distribution in $U[0, 1]$. The corresponding sampled class vectors are shown above each plot, with the dominant glitch label also shown. Both methods produce hybrid morphologies interpolating between glitch classes, demonstrating GlitchGAN’s ability to explore smooth transitions across the learned latent class space.

diffusion is a viable approach to conditional time-domain glitch generation, its ability to reproduce full morphological diversity—especially for mid- and high-frequency content—is limited relative to GlitchGAN.

V. DISCUSSION

A. Performance assessment and limitations

GlitchGAN successfully demonstrates time-domain synthesis of multiple realistic glitch classes. Substantial UMAP overlap between real and synthetic samples, combined with accurate Gravity Spy classification of generated glitches indicates morphological fidelity of generated glitches. However, there exists several limitations and caveats which we address here.

Partial failure was observed for the Whistle class, possibly due to insufficient regularization or architectural constraints on receptive field size. For example, smaller kernel sizes in generator and discriminator layers may be

needed to capture the high-frequency content of Whistle glitches (please see Appendix A for details).

Since GlitchGAN is trained on normalized waveforms, it does not learn an explicit SNR distribution, and thus generated glitches must be rescaled to a specified SNR for use in downstream applications. As shown in Section IV B 1, Gravity Spy classifications are strongly sensitive to the injected SNR — consistent with the fact that individual glitch classes are partly characterized by their SNR distribution. Appropriate rescaling is therefore an important consideration for simulations and pipeline validation studies using these synthetic samples. For example, one may choose to rescale generated glitches to match the SNR distribution of the real glitches within a given class, or to a fixed SNR for specific applications such as testing pipeline sensitivity at particular SNR thresholds.

Finally, we note that no hyperparameter optimization was performed for this specific dataset; the model is evaluated “out-of-the-box” from previous work (except for the dimensionality of the class vector), representing a

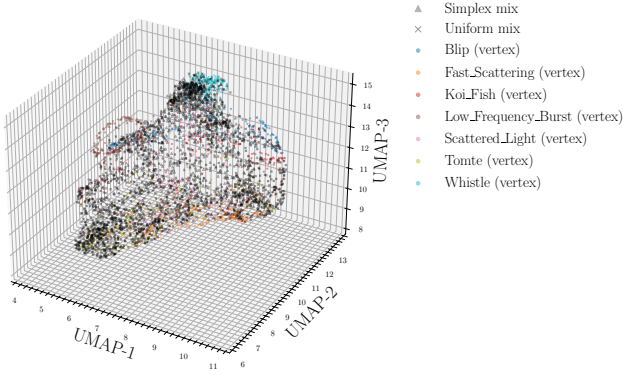


FIG. 8: Three-dimensional UMAP embedding showing the distribution of class-mixed glitch samples relative to the vertex (pure-class) glitches. Vertex samples are coloured by their respective glitch class, while class-mixed samples generated using simplex and uniform class-conditioning vectors are shown in black. The smooth interpolation of mixed samples between class clusters illustrates how GlitchGAN captures continuous transitions across glitch morphologies in the latent class space.

conservative assessment of GlitchGAN’s potential. Tuning the parameters of the generator and discriminator architectures, as well as the training procedure, could further improve performance, particularly for the more challenging Whistle class.

B. Practical applications

The synthetic glitch population produced by GlitchGAN is immediately useful for:

1. **Mock data challenges (MDCs):** Enriching realistic detector simulations with synthetic instrumental noise
2. **Detector characterization:** Validating glitch detection, mitigation, and subtraction algorithms
3. **Training data augmentation:** Supplementing limited real glitch samples with GlitchGAN-generated samples
4. **Pipeline robustness testing:** Assessing pipeline sensitivity and false-alarm rates in the presence of instrumental noise morphologies

C. Recommendations for future work

There are several promising directions for future research building on this work. Firstly, hyperparameter optimization and architectural refinements could be performed to further improve the quality of generated data, particularly for the more challenging Whistle class. Architecture improvements may include the incorporation

of attention mechanisms, smaller convolutional kernel sizes or dilated convolutions for multi-scale feature extraction.

Ultimately, the goal is to develop a comprehensive generative model capable of synthesizing the entire range of glitch morphologies in GW detectors with high fidelity, enabling robust simulations and pipeline validation for current and future GW detectors. Future work will iteratively include more glitch glitches, starting with more common and progressively incorporating less common ones.

It would be beneficial to develop more sophisticated validation metrics that explicitly evaluate phase consistency, not just magnitude-level morphology, to ensure synthetic glitches are suitable for detector simulations. Such methods will be beneficial for evaluating the realism of synthetic glitches and guiding future improvements in generative modeling approaches.

While class conditional generation is a powerful tool, it may also be desirable to develop conditional generation strategies allowing control over glitch duration, SNR, and frequency content independently from class morphology. This would enable more flexible synthesis of glitches for specific applications, such as testing pipeline sensitivity to particular frequency bands or durations.

Finally, integrating these features with mock data challenge pipelines for rapid deployment in detector simulations will be an important step towards practical utility of these synthetic glitch populations for the GW data analysis community.

VI. CONCLUSIONS

We have demonstrated that a class-conditional deep generative model called GlitchGAN can effectively synthesize realistic GW detector glitches directly in the time domain. By leveraging high-fidelity reconstructions from a glitch reconstruction framework called DeepExtractor and training a single unified GlitchGAN model based on the cDVGAN architecture across seven glitch classes, we achieve synthetic samples with strong morphological agreement with real data. GlitchGAN’s synthetic glitch population offers practical utility for mock data challenges, detector characterization studies, and validation of detection pipelines.

We evaluated both DeepExtractor’s reconstructions used to train GlitchGAN and GlitchGAN’s synthetic data using two complementary approaches: a supervised classification analysis using Gravity Spy, a state-of-the-art glitch classification algorithm trained on an extensive labeled dataset, and an unsupervised, data-driven UMAP analysis to visualize and compare the class distributions of real and synthetic glitches in a projected three-dimensional space.

The Gravity Spy classification results for DeepExtractor reconstructions showed that 89.57% of the reconstructed glitches were correctly classified, with a

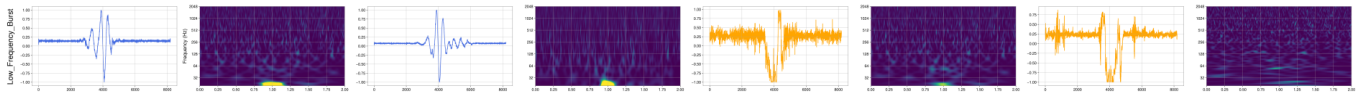


FIG. 9: Two examples each of real (left) and DiT-generated (right) Low Frequency Burst glitches, showing both the time-domain waveform and the magnitude Q -scan representation. The time-domain waveforms are clearly distinct, yet the Q -scans appear similar, explaining why Gravity Spy classifies the DiT samples correctly with high confidence.

mean classifier confidence of 96.77% for correct predictions. Misclassified samples showed a lower confidence of 84.84%, suggesting that classifier confidence can serve as a quality veto. Overall, the misclassification patterns are generally explainable, typically misclassifying as morphologically similar classes, and do not indicate systematic failure of the reconstruction. The unsupervised UMAP analysis of DeepExtractor reconstructions revealed well-separated clusters according to the Gravity Spy labels, demonstrating that the reconstructions preserve class-specific morphological structure in the time domain.

For GlitchGAN’s generated glitches, Gravity Spy classified most samples correctly with an overall accuracy of 79%. Misclassifications primarily occurred for the Koi Fish and Scattered Light classes, which can largely be attributed to morphological similarities between similar classes and Gravity Spy’s classification being dependent on SNR. For instance, Scattered Light glitches were often misclassified as Fast Scattering, while Koi Fish are often misclassified as Extremely Loud or Light Modulation. These overlaps are expected, as the classes share similar time-frequency signatures. This is confirmed by changing the injected SNR to 150 for Koi Fish samples and 50 for the Whistle class, where 79% and 94% are correctly classified, respectively.

The UMAP results showed that GlitchGAN-generated glitches generally occupy the same regions as real data in the embedded UMAP space, indicating that the model captures much of the morphological diversity within each glitch class. One exception is the Whistle class, which consists of the highest-frequency features in the dataset. For this class, synthetic samples formed a partially distinct cluster from the real data, suggesting that GlitchGAN did not fully capture the fine-scale, high-frequency structure characteristic of these glitches. Future work could address this by incorporating smaller convolutional kernels in the network (balanced against the need for long-range temporal modelling), dilated convolutions to better capture hierarchical structure, or attention mechanisms to enhance global feature representation.

Finally, this study highlights a fundamental limitation in using purely magnitude-based Q -scan representations for supervised glitch classification. We observed that a less robust diffusion-based model, which produces unrealistic Low Frequency Burst glitches in the time domain, can still yield correct Gravity Spy labels with high confidence. Although the time-domain waveforms of real and synthetic samples are clearly distinct, their Q -scan representations appear similar, explaining the classifier’s high confidence. This discrepancy is evident in the cor-

responding UMAP analysis of time-domain data, which reveals two distinct clusters separating real and synthetic samples. This finding highlights the importance of preserving phase information and integrating unsupervised analyses to complement glitch characterization efforts.

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Appendix A: cDVGAN Architecture

Table V shows the cDVGAN architecture featured in GlitchGAN.

		Discriminator (3.5M parameters)			
Operation	Output shape	Kernel size	Stride	Dropout	Activation
Input	(1024)	–	–	0	–
Reshape	(64,16)	–	–	0	–
Convolutional	(64,128)	14	2	0.5	Leaky ReLU
Convolutional	(32,128)	14	2	0.5	Leaky ReLU
Convolutional	(16,256)	14	2	0.5	Leaky ReLU
Convolutional	(8,256)	14	2	0.5	Leaky ReLU
Convolutional	(4,512)	14	2	0.5	Leaky ReLU
Global Avg. Pooling	(512)	–	–	0.5	–
Avg. Pooling Dense	(128)	–	–	0.2	Leaky ReLU
Dense	(1)	–	–	0	Linear
Class input	(7)	–	–	–	–
Class Dense	(128)	–	–	0	Linear
Scalar product	(1)	–	–	–	–
Dense + Scalar product	(1)	–	–	–	–
		DV Discriminator (1.1M parameters)			
Operation	Output shape	Kernel size	Stride	Dropout	Activation
Input	(1023)	–	–	0	–
Dense	(512)	–	–	–	Leaky ReLU
Reshape	(32,16)	–	–	0	–
Convolutional	(32,64)	5	2	0.5	Leaky ReLU
Convolutional	(16,128)	5	2	0.5	Leaky ReLU
Convolutional	(8,256)	5	2	0.5	Leaky ReLU
Convolutional	(4,256)	5	2	0.5	Leaky ReLU
Global Avg. Pooling	(256)	–	–	0.5	–
Avg. Pooling Dense	(128)	–	–	0.2	Leaky ReLU
Dense	(1)	–	–	0	Linear
Class input	(7)	–	–	–	–
Class Dense	(128)	–	–	0	Linear
Scalar product	(1)	–	–	–	–
Dense + Scalar product	(1)	–	–	–	–
		Generator (3.5M parameters)			
Operation	Output shape	Kernel size	Stride	BN	Activation
Latent input	(100)	–	–	–	–
Class input	(7)	–	–	×	–
Class Dense	(32)	–	–	×	–
Concatenate	(132)	–	–	×	–
Dense	(1024)	–	–	×	ReLU
Reshape	(32,32)	–	–	×	–
Transposed conv.	(64,512)	18	2	✓	ReLU
Transposed conv.	(128,256)	18	2	✓	ReLU
Transposed conv.	(256,128)	18	2	✓	ReLU
Transposed conv.	(512,64)	18	2	✓	ReLU
Transposed conv.	(1024,1)	18	2	×	Linear
Flatten	(1024)	–	–	×	–
Optimizer	RMSprop ($\alpha = 0.0001$)				
Batch size	512				
Epochs	500				
Loss	Wasserstein				

TABLE V: Architecture and hyperparameters of cDVGAN featured in GlitchGAN. The model includes a base discriminator, derivative (DV) discriminator, and generator.

Appendix B: cDVGAN Loss

Figure 10 shows the training loss for each individual model component.

Appendix C: Hyperparameters for t-SNE and UMAP

Table VI lists the hyperparameters used for the t-SNE and UMAP analyses.

TABLE VI: Hyperparameters for t-SNE and UMAP

Algorithm	Hyperparameter	Value
t-SNE	N_components	3
	Perplexity	40
	Metric	euclidean
UMAP	N_neighbors	15
	Min_dist	0.6
	N_components	3
	Metric	correlation

Appendix D: DiT evaluation results

Figures 11 and 12 show the full UMAP and Gravity Spy evaluation results for the DiT comparison model.

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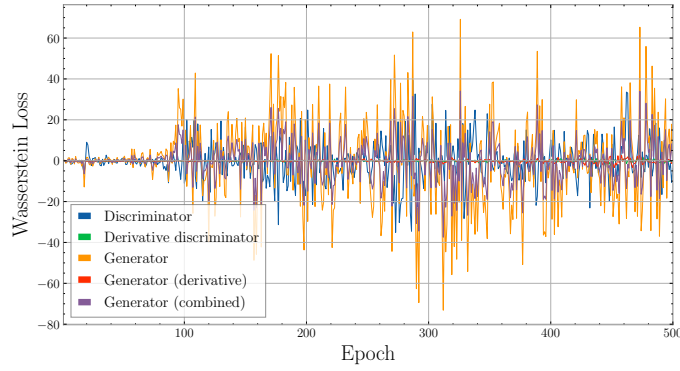


FIG. 10: *GlitchGAN* training loss over 500 epochs.

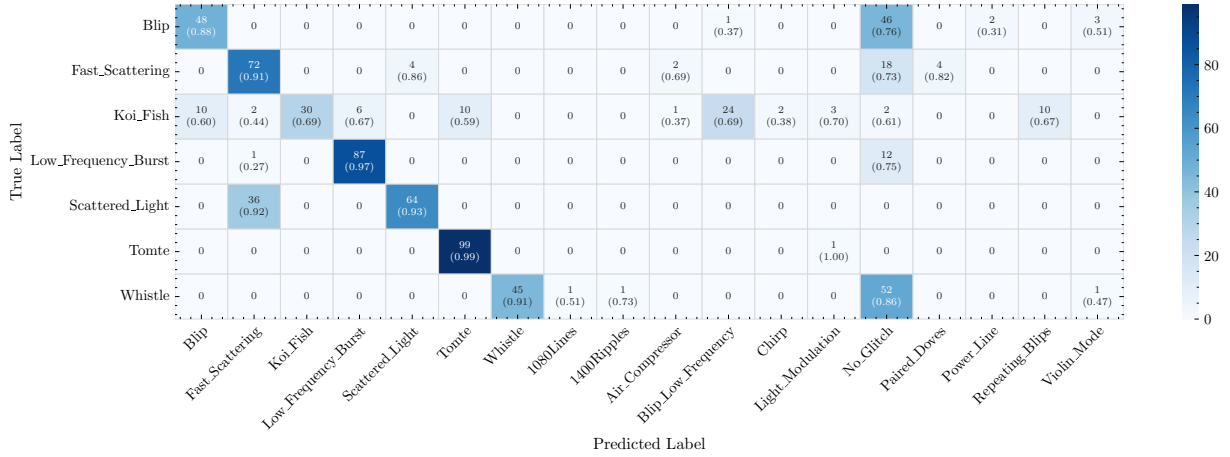


FIG. 11: *Gravity Spy* confusion matrix for samples generated by the DiT model (100 per class). Each cell shows the number of predictions and mean classifier confidence in brackets. Despite the poor UMAP overlap, *Gravity Spy* achieves high confidence classification for most classes, illustrating the limitation of magnitude-only validation.

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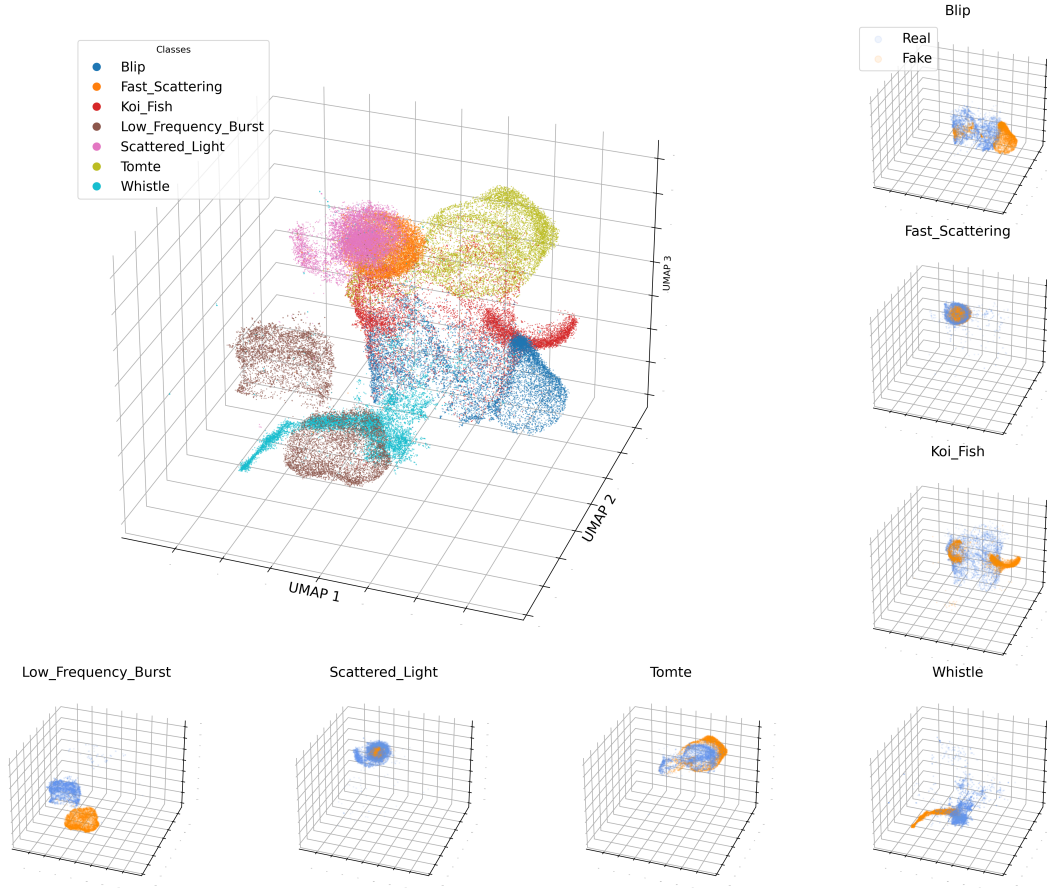


FIG. 12: Three-dimensional UMAP projection of real (orange) and DiT-generated (blue) glitches across all seven classes. The large panel shows the overall structure, while smaller panels display each class individually. Overlapping clusters indicate morphological similarity between real and synthetic samples; separable clusters for most classes indicate that DiT does not reproduce the full morphological structure of real glitches, in contrast to GlitchGAN (Figure 5).

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